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Research Article

Received: 27 April 2025 Revised: 6 November 2025 Published online in Wiley Online Library: 22 November 2025

(wileyonlinelibrary.com) DOI 10.1002/ps.70393

Predicting wheat powdery mildew epidemics in China using meteorological data and machine learning approaches

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Abstract

BACKGROUND: Prediction is vital for plant disease management. This study developed machine learning models that used meteorological data to predict wheat powdery mildew (WPM) occurrence severity degree and area in China. Six machine learning algorithms were trained and cross-validated to predict WPM severity degree with 411 pieces of meteorological data from 48 counties (1981–2021) across China. Areas of WPM occurrence were also derived from WPM severity degrees [which were predicted by the K-Nearest Neighbor (KNN) model] with spatial interpolation models.

RESULTS: The best-performing machine learning severity prediction models were based on meteorological data during the coldest month (January) of the wheat overwintering period, and also the wheat jointing stage–heading stage. In each case the times were subdivided into 5-day periods. In particular, the prediction model showed that the best performance was based on the support vector machine algorithm. Climate variable importance ranked via random forest identified eight key predictors. Using these, KNN achieved high performance, demonstrating its suitability for predicting WPM severity degree. Nationwide severity distributions were produced using inverse distance weighted (IDW) and ordinary kriging methods, based on severity degrees predicted by the KNN model from 1990 to 2019. Validation via chi-squared and error reference methods confirmed that the IDW_4.0 model outperformed the others.

CONCLUSIONS: Machine learning models effectively predict WPM severity degree and area of occurrence at a national scale using meteorological data. The disease severity distribution of WPM displays disease severity spatial patterns visually and can improve management strategies for WPM across China.
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Supporting information may be found in the online version of this article.

Keywords: *Blumeria graminis* f. sp. *tritici*; plant disease prediction models; spatial interpolation; artificial intelligence; climate variables

1 INTRODUCTION

Wheat powdery mildew (WPM), caused by *Blumeria graminis* f. sp. *tritici* (*Bgt*), is one of the most important diseases of wheat in worldwide.[1–3] It can lead to estimated annual yield losses of 7.9–19.9% on a global scale.[4–6] The average occurrence area of WPM is 6–10 million hectares,[7] causing 10–30% yield losses annually in China.[8]

Accurate prediction plays a crucial role in disease management. Many statistical models have been established for predicting WPM epidemics using meteorological data in Jiangsu,[9] Sichuan,[10] and Hebei[11] provinces in China. Empirical models using meteorological data (temperature, humidity and rain in April to June among others) were formulated and used to predict WPM epidemics quantitatively and qualitatively (>5% severity) in the UK.[12] Nie *et al.* collected and compared 46 statistical models for predicting WPM across 31 cities or counties in 9 provinces (autonomous regions, municipalities) in China, and found that all these statistical models had limited timeliness and narrow prediction ranges.[13]

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Therefore, considering the model's generality, machine learning models have demonstrated broader performance with a larger scope than statistical models.[14,15] For instance, De-Wolf *et al.* developed a neural networks model to predict epidemics of wheat leaf spot using meteorological data, and the accuracy and prediction range were higher than that of stepwise logistic regression models.[16] Shah *et al.* investigated the utility of boosted regression trees for predicting the epidemic of Fusarium head blight in the United States, and the results showed that models built using boosted regression trees had a better accuracy and prediction range than logistic regression models.[17]

No WPM prediction models based on machine learning algorithms that use meteorological data from China have been reported as yet. The objective of this study was to establish and select an optimal prediction model for WPM severity degree with nationwide generalizability based on six machine learning algorithms, utilizing nearly four decades (1981–2021) of disease severity degree and corresponding meteorological variable data sets from major winter wheat regions in China. In addition, predicted disease severity degrees and spatial interpolation methods will be used to produce a national distribution of WPM severity. Disease occurrence area for each location will be subsequently predicted by the spatial distribution qualitatively and visually. The results of this study are intended to provide a scientific reference for guiding the prevention and control of WPM.

2 MATERIALS AND METHODS

2.1 Disease and meteorological data

2.1.1 *WPM severity data*

WPM severity data (site years) were collected from 48 counties (or cities) in 15 provinces in China, covering 1981–2021, using sources comprising published papers, the plant protection stations of different provinces, and the National Agro-Tech Extension and Service Center (NATESC) (Supporting Information, Table S1). The 411 WPM severity data sets were handled using the synthetic minority oversampling technique (SMOTE) by function in imblearn.over_sampling 0.0 in Python 3.9.[18]

The collected WPM severity data consist of in six classifications: 0, zero occurrence [disease index (DI) = 0]; 1, low occurrence ($0 < DI \leq 10$); 2, low–moderate occurrence ($10 < DI \leq 20$); 3, moderate occurrence ($20 < DI \leq 30$); 4, moderate–high occurrence ($30 < DI \leq 40$); and 5, high occurrence ($DI > 40$). This classification was based on the range of the DI values, according to National Standard of China rules (NY/T 613–2002) for investigating and forecasting WPM. To reduce data complexity, improve model training and facilitate comparison between models, the WPM severity data were categorized as three epidemic degrees (WPM severity degree): L, none or low epidemic; M, moderate epidemic; and H, heavy epidemic (Supporting Information, Table S2).

2.1.2 *Meteorological data*

Meteorological data for building the WPM severity degree prediction models included daily mean temperature (°C), precipitation (mm), relative humidity (%) and sunshine hours (h) from 692 stations across China during 1981–2021, and were obtained from the National Meteorological Information Center (http://data.cma.cn/site/index.html) (Supporting Information, Table S3).

Previous research showed WPM epidemics were associated with meteorological conditions during the winter and early spring in China.[19–23] Therefore, we chose the meteorological data for these two critical periods for model development: (i) the coldest month from 1 to 31 January, corresponding to wheat tillering to erect-leaf sheath stage growth stage (GS)[24] 3–4; and (ii) 1 month before the wheat jointing or heading stage in early spring, corresponding to GS 6 to GS 10.1.

Climate variables from the coldest month (1–31 January; GS 3–GS 4) were computed forwards in 5-day (X_1–X_{24}) and 10-day (X_{25}–X_{36}) intervals (Supporting Information, Table S3). Pre-heading stage variables (GS 6–GS 10.1) were calculated backward in 5-day (X_{37}–X_{60}) and 10-day (X_{61}–X_{72}) intervals before heading, with earlier periods representing longer pre-heading durations (Supporting Information, Table S3).

2.2 Machine learning algorithms

The workflow for predicting WPM epidemics based on machine learning algorithms and meteorological data in China is shown in Fig. 1(a). Six machine learning algorithms were selected: K-Nearest Neighbor (KNN),[25] decision tree (DT),[26,27] random forests (RF),[28,29] support vector machine (SVM),[30,31] adaptive boosting (AdaBoost),[32] and artificial neural networks (ANN).[33] Each model was trained and cross-validated using WPM severity degrees and meteorological data.

The six input data sets comprised WPM severity degrees and meteorological data: data set 1, coldest month (January, GS 3–GS 4) in 5-day intervals (X_1–X_{24}) and associated WPM severity degrees; data set 2, pre-heading month (GS 6–GS 10.1) in 5-day intervals (X_{37}–X_{60}) and associated WPM severity degrees; data set 3, combined all data from data set 1 and data set 2; data set 4, coldest month (January, GS 3–GS 4) in 10-day intervals (X_{25}–X_{36}) and associated WPM severity degrees; data set 5, pre-heading month (GS 6–GS 10.1) in 10-day intervals (X_{61}–X_{72}) and associated WPM severity degrees; and data set 6, combined all data from data set 4 and data set 5. All meteorological data were standardized by the zero-mean normalization method with Python before developing each model (The disease severity degrees were not standardized). The six machine learning algorithms were implemented with the Python package scikit-learn (0.24.2). The prediction accuracy of the machine learning algorithms was mainly affected by the choice of parameters.

Each machine learning algorithm has a set of hyper-parameters. Some algorithms were optimized by an exhaustive search of a network of hyper-parameter values (Table 1); others were kept to their recommended values (hyper-parameters not listed were set to their default values in scikit-learn).

2.3 Climate variable importance, selection and validation

Figure 1(a) shows the workflow for the variable's importance, selection and validation. Variable importance was calculated using the RF algorithm based on the 'mean decrease accuracy' metric.[15,34] This method permutes each variable in out-of-bag samples while keeping the importance of the variable unchanged, and measures the corresponding decrease in the model's prediction accuracy. The average reduction in accuracy across all trees in the forest was used to quantify the importance of each variable.

Subsequently, variables were ranked according to their importance scores. To identify the most predictive and parsimonious data set for meteorological data, four distinct data sets (numbers 7–10) were constructed comprising the top four, six, eight and ten most important variables and associated WPM severity degrees, respectively. The prediction performances of models built on these four data sets were then rigorously evaluated and

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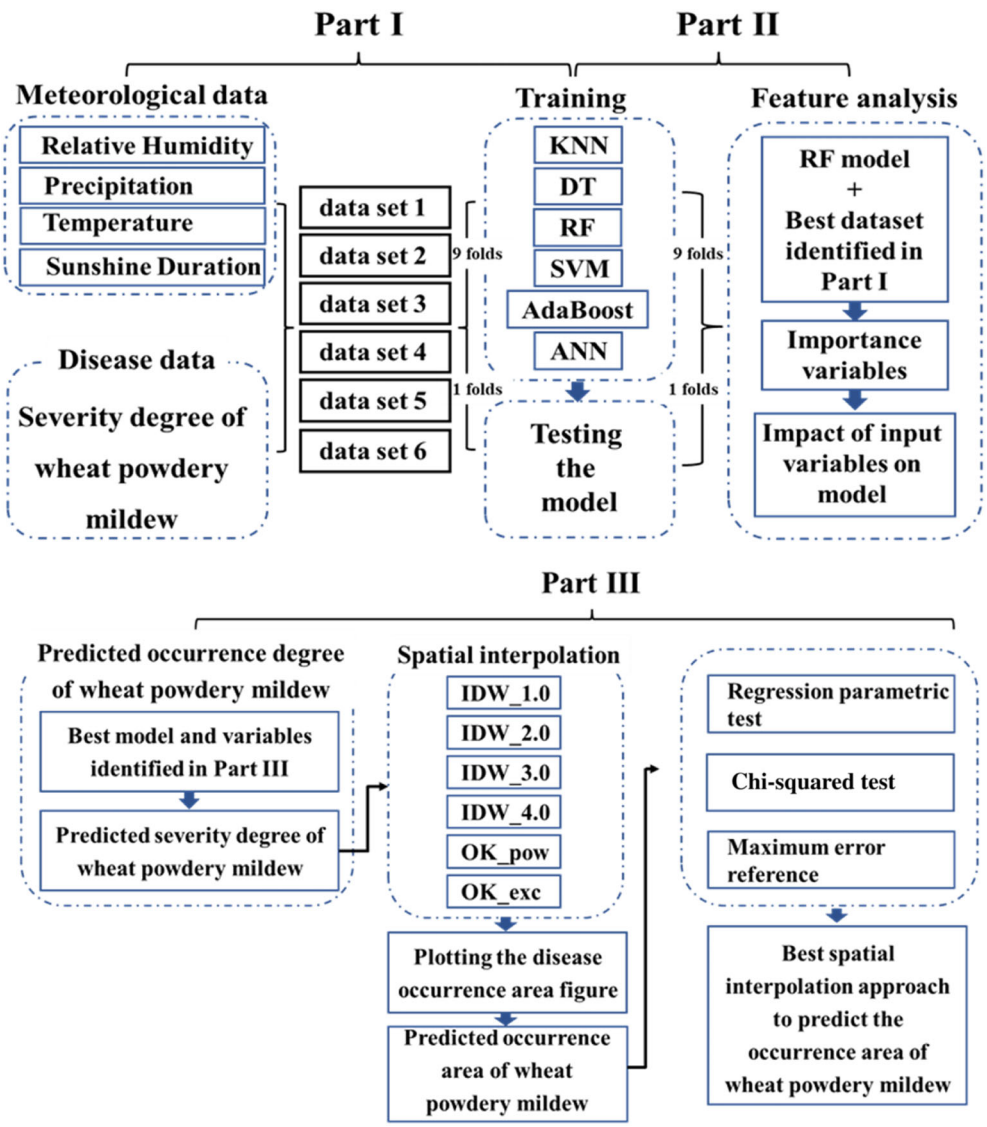


Figure 1. Workflow for predicting wheat powdery mildew epidemics in China using machine learning algorithms and meteorological data. (a) Select the best-performing prediction model from the full, unfiltered data set and its suitable machine learning algorithm. (b) Refine the data set by selecting key meteorological variables to build an optimized prediction model using the simplified data set. (c) Use the optimized model and data set together with spatial interpolation methods to predict the occurrence area of wheat powdery mildew. KNN, K-Nearest Neighbor; DT, decision tree; RF, random forest; AdaBoost, adaptive boosting; SVM, support vector machine; ANN, artificial neural network; IDW, inverse distance weighted; OK, ordinary kriging. Data set 1: coldest month (Jan, GS 3–GS 4) in 5-day intervals (X_1–X_{24}) and associated wheat powdery mildew (WPM) severity degrees; data set 2: pre-heading month (GS 6–GS 10.1) in 5-day intervals (X_{37}–X_{60}) and associated WPM severity degrees; data set 3: combined all data from data set 1 and data set 2; data set 4: coldest month (Jan, GS 3–GS 4) in 10-day intervals (X_{25}–X_{36}) and associated WPM severity degrees; data set 5: pre-heading month (GS 6–GS 10.1) in 10-day intervals (X_{61}–X_{72}) and associated WPM severity degrees; data set 6: combined all data from data set 4 and data set 5.

compared (Section 2.4). The optimal data set was selected based on this model's performance evaluation, which aimed to balance model complexity with prediction performance.

2.4 Model evaluations and validation

Model performances were evaluated by classification accuracy, precision, recall and F1-score using the Python package scikit-learn. Classification accuracy measures overall correctness, whereas precision focuses on prediction reliability among positives. Recall assesses the detection of all actual positives, and the F1-score balances the latter two, providing a single metric in which both false positives and negatives are important.

The model performances were calculated as:

$$\text{Classification accuracy} = \frac{TP+TN}{TP+FP+FN+TN}$$

$$\text{Precision} = TP/(TP+FP)$$

$$\text{Recall} = TP/(TP+FN)$$

$$\text{F1-score} = 2\times\left(\frac{\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}}\right)$$

where the true positive (TP) is the number of times that the WPM severity degrees were predicted to be positive and were actually

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Table 1. List of key values of hyper-parameters for the six machine learning algorithms

Machine learning algorithm	Hyper-parameter	Values of hyper-parameter
KNN	K	1
DT	max_depth	10
	min_samples_leaf	1
	criterion	entropy
RF	n_estimators	171
	max_depth	16
	max_features	2
	min_samples_leaf	1
	min_samples_split	2
	criterion	gini
SVM	C	30
	gamma	0.019
	kernel	rbf
AdaBoost	max_depth	10
	min_samples_leaf	3
	min_samples_split	0
	n_estimators	55
	learning_rate	0.6
	criterion	entropy
ANN	hidden_layer_sizes	50, 50, 50
	solver	adam
	activation	relu
	alpha	0.05

AdaBoost, adaptive boosting; ANN, artificial neural network; DT, decision tree; KNN, K-Nearest Neighbor; RF, random forest; SVM, support vector machine.

positive; false positive (FP) is the number of times that the WPM severity degrees were predicted to be positive but were actually negative; false negative (FN) is the number of times that the WPM severity degrees were predicted to be negative but were actually positive; and true negative (TN) is the number of times that the WPM severity degrees were predicted to be negative and were actually negative.

$$\text{Macro}-\text{precision}=\frac{P_L+P_M+P_H}{3}$$

where P_L is the precision of class 'L', P_M is the precision of class 'M' and P_H is the precision of class 'H'.

$$\text{Macro}-\text{recall}=\frac{R_L+R_M+R_H}{3}$$

where R_L is the recall of class 'L', R_M is the recall of class 'M' and R_H is the recall of class 'H'.

$$\text{Macro F1}-\text{score}=\frac{F_L+F_M+F_H}{3}$$

where F_L is the F1-score of class 'L', F_M is the F1-score of class 'M', and F_H is the F1-score of class 'H'.

To evaluate model performance, tenfold cross-validation was applied via sklearn.model_selection.cross_val_score.[35] The data set was randomly divided into ten subsets, with each iteration using nine subsets for training and one subset for validating—ensuring complete separation between model development and validation. This process was repeated until all subsets served as the test set once. Final accuracy and macro-precision metrics represent average values across all iterations, reflecting the models' generalized performance and applicability.

2.5 Predicting the occurrence area of WPM

The predicted WPM severity degree was calculated using a model based on a certain machine learning algorithm and certain data set during 1990 to 2019 with the Python package scikit-learn. WPM severity distributions were produced using the predicted WPM severity degrees and spatial interpolation methods. Based on these distributions, the predicted WPM occurrence rate was calculated, and combined with wheat planting area data to predict the WPM occurrence area (Fig. 1(c)).

WPM occurrence area data for China from 1990 to 2019 were sourced from NATESC. Wheat planting area data for all provinces (including municipalities and autonomous regions) across China from 1990 to 2019 were obtained from the National Bureau of Statistics of China (https://data.stats.gov.cn/). Meteorological data from 1990 to 2019 were used to predict WPM occurrence area (Section 2.1.2).

For predicting the WPM occurrence area, inverse distance weighted (IDW) and ordinary kriging (OK) interpolation were used in R 4.1.2.[36,37] In IDW, the power function (p) controls the significance of the known points in the interpolated values based on proximity, and therefore was defined as p = 1.0, 2.0, 3.0 or 4.0. In OK, exponential class and power models were used.

The predicted WPM occurrence area in China for a given year was calculated as the sum of the predicted WPM area in each province, weighted by the corresponding winter wheat planting area:

$$OA=\sum_{i=1}^{n} A_i \times a_i$$

For a given year, where OA is the total predicted WPM occurrence area in China; A_i is winter wheat planting area of province 'i'; and a_i is the predicted WPM occurrence rate for province 'i'.

a_i was calculated as:

$$a_i=\frac{\text{Number of occurrence area pixels in province } i'}{\text{Total plnting area pixels in province } i'}$$

2.6 Validating the predicted WPM occurrence area

The accuracies of the predicted WPM occurrence areas were validated using the chi-squared test[38] and maximum error reference methods[39] in R 4.1.2.

Chi-squared test used to validate the fit of the model was calculated as:

$$\chi^2=\sum_{i=1}^{n}\frac{(A-T)^2}{T}$$

The maximum error reference method used to validate the relative performance of the different models was calculated as:

$$\text{Accuracy}=\frac{1}{n}\sum_{i=1}^{n}\left(1-\frac{|F_i-T_i|}{M_i}\right)\times 100\%$$

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Where χ^2 is the value of the chi-squared test; Accuracy is the accuracy for predicting WPM occurrence area; A is the actual WPM occurrence area and T is the predicted WPM occurrence area; F_i is the actual occurrence area of WPM for i observations; T_i is the predicted occurrence area of WPM for the ith observation; and M_i is the maximum reference error for the ith prediction, which is the largest of the actual occurrence area and the difference between the actual maximum occurrence area and the actual occurrence area.

3 RESULTS

3.1 Construction and comparison of prediction models for WPM severity degree

Six machine learning prediction models for WPM severity degree were developed based on data sets 1 to 6. The prediction accuracy and macro-precision of these prediction models were obtained through a cross-validation method (Figs 2 and 3). Regarding accuracy, the SVM model demonstrated strong overall performance on data sets 1 to 6 by achieving the highest accuracy on data set 2 (0.75), data set 3 (0.76), data set 4 (0.74), data set 5 (0.70) and data set 6 (0.75), and the second-highest accuracy on data set 1(0.71). The KNN model also performed exceptionally well, achieving the highest accuracy on data set 1 (0.72), the second-highest accuracy on data set 2 (0.74) and data set 3 (0.75), and it also performed relatively well on data sets 4–6 (0.68–0.71). Conversely, the DT model consistently delivered the poorest accuracy across all data sets (0.55–0.60), indicating it was the least applicable model. The RF, AdaBoost and ANN models achieved accuracies ranging from 0.64 to 0.71, 0.56 to 0.69 and 0.67 to 0.73 across all data sets, respectively. Overall, SVM and KNN were superior to the other four models in terms of prediction accuracy.

In terms of macro-precision, the ANN model achieved highest macro-precision on data set 1 (0.87) and data set 4 (0.77), and the second-highest macro-precision on data set 6 (0.69). The SVM model demonstrated relatively high macro-precision across all six data sets (0.63–0.81), it achieved the highest macro-precision on data set 3 (0.81), data set 5 (0.78), and data set 6 (0.74), the second-highest macro-precision on data set 2 (0.73). The KNN model achieved highest macro-precision on data set 3 (0.86), and the macro-precision on the other 5 data sets ranged from 0.55 to 0.72. AdaBoost demonstrated very high macro-precision on data set 2 (0.76) and data set 4 (0.76), but its poor performance on data set 3 (0.51, the worst among all models for data set 3) reveals critical stability issues with the model. The RF and DT models exhibited relatively low macro-precision across all six data sets, and the DT model performed particularly poorly on data set 4 (0.43). Overall, SVM, ANN and KNN were superior to the other three models in terms of macro-precision.

Balancing stable accuracy and standout macro-precision, SVM and KNN can be considered the optimal algorithms for predicting WPM severity degree in this study. They maintained consistently higher performance across both accuracy and macro-precision, demonstrating superior robustness.

From the data set perspective, data set 3 facilitated the highest accuracy for SVM (0.76) and exceptionally high macro-precision for KNN (0.86). This suggests that the meteorological data contained in data set 3 have the strongest correlation with disease severity degree and the highest informational value, making data set 3 the recommended data set.

Consequently, the initial analysis incorporating all meteorological data indicated that the SVM and KNN models delivered optimal WPM prediction performance on data set 3.

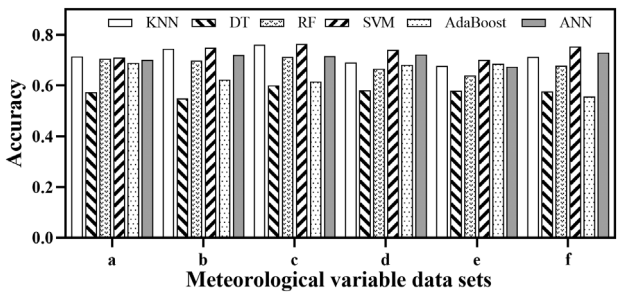


Figure 2. The accuracies of wheat powdery mildew severity degree prediction in China using six machine learning algorithms based on data sets 1–6. a, b, c, d, e and f represent data sets 1, 2, 3, 4, 5 and 6, respectively.

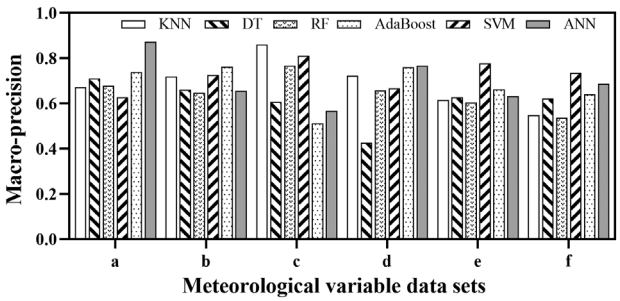


Figure 3. The macro-precisions of wheat powdery mildew severity degree prediction in China using six machine learning algorithms based on data sets 1–6. a, b, c, d, e and f represent data sets 1, 2, 3, 4, 5 and 6, respectively.

3.2 Climate variable importance, selection and validation

Dimensionality reduction was performed on meteorological data set 3 (coldest and pre-heading month in 5-day intervals), and the relative importance of the top ten climate variables used with data set 3 based on RF by ranking the variables (Table S3) are shown Fig. 4. The mean temperature for days 6–10 before the heading stage (GS 6–GS 10.1) (X_{41}), mean sunshine duration for days 1–5 in January (GS 3–GS 4) (X_4), mean temperature for days

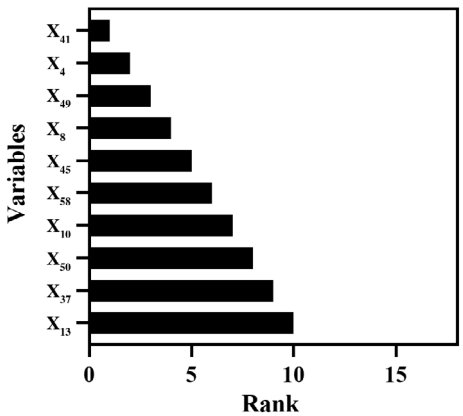


Figure 4. The top ten important climate variables based on random forest. Climate variables are listed in Supporting Information, Table S3.

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16–20 before the heading stage (GS 6–GS 10.1) (X_{49}), mean sunshine duration for days 6–10 in January (GS 3–GS 4) (X_8), mean temperature for days 11–15 before the heading stage (GS 6–GS 10.1) (X_{45}), mean relative humidity for days 26–30 before the heading stage (GS 6–GS 10.1) (X_{58}), mean relative humidity for days 11–15 in January (GS 3–GS 4) (X_{10}), mean relative humidity for days 16–20 before the heading stage (GS 6–GS 10.1) (X_{50}), mean temperature for days 1–5 before the heading stage (GS 6–GS 10.1) (X_{37}) and mean temperature for days 16–20 in January (GS 3–GS 4) (X_{13}) were the top ten most important climate variables for WPM severity degree prediction.

Four new data sets containing the top four (data set 7), six (data set 8), eight (data set 9) and ten (data set 10) most influential variables and associated WPM severity degrees, were established based on variable ranking and dimensionality reduction on meteorological data set 3, and the prediction performance of these four meteorological data sets (7–10) also were validated using prediction accuracy and macro-precision assessments across six machine learning algorithms (Figs 5 and 6).

In terms of accuracy, the KNN model showed highest accuracy using data set 9 (0.68) and data set 10 (0.71). The RF model showed the highest accuracy using data set 7 (0.60). The ANN model showed highest accuracy using data set 8 (0.66). The SVM model showed second-highest accuracy using data set 9 (0.67) and data set 10 (0.70). The other models achieved lower accuracy (0.51–0.66). In terms of macro-precision, the KNN model showed highest macro-precision (0.82) using data set 9. The SVM model showed highest macro-precision (0.76) using data set 10. The RF model showed highest macro-precision using data set 7 (0.73) and data set 8 (0.66). The AdaBoost model showed second-highest macro-precision using data set 9 (0.68). The other models achieved lower macro-precision (0.46–0.62).

Considering both predictive efficiency and model parsimony, the KNN model with data set 9 was selected for WPM severity degree prediction because of its balanced accuracy (0.68), precision (0.82) and minimal variable requirements.

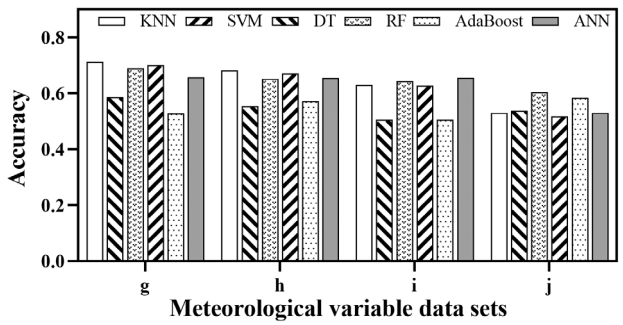


Figure 5. The accuracies of wheat powdery mildew severity degree prediction in China using data sets 7–10. g, h, i and j represent results using the top ten (data set 10), top eight (data set 9), top six (data set 8) and top four (data set 7) most influential variables and their disease severity degree, respectively.

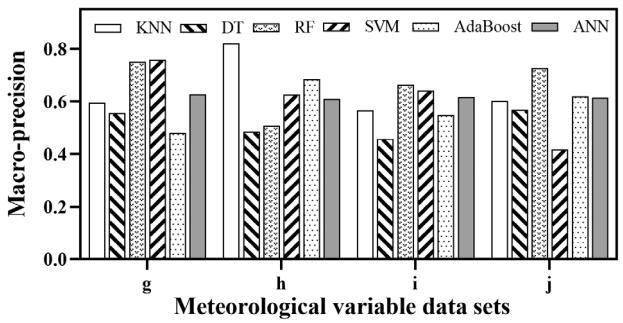


Figure 6. The macro-precisions of wheat powdery mildew severity degree prediction in China using data sets 7–10. g, h, i and j represent results using the top ten (data set 10), top eight (data set 9), top six (data set 8) and the top 4 (data set 7) most influential variables and their disease severity degree, respectively.

3.3 Predicting the occurrence area of WPM in China

The WPM-predicted severity degrees were calculated based on the KNN algorithm with data set 9 and predicted disease distributions of WPM were produced with predicted severity degrees and different spatial interpolation models. Figure 7 shows examples of the predicted WPM distributions throughout China for two different epidemics years: a heavy epidemic in 1991 and light epidemic in 2002, as a spatial interpolated method using IDW_4.0. Table 2 shows the predicted and actual WPM occurrence areas throughout China during 1990–2019. The results showed that predicted and actual WPM occurrence areas were consistent with almost all six spatial interpolation methods, but there were also differences in some years. Overall, this result suggests the potential of using machine learning and spatial interpolation models to predict WPM occurrence area at the regional level.

3.4 Validating the WPM occurrence area in China

To compare the performance of the spatial interpolation models, we validated their predicted WPM occurrence areas against the actual occurrence areas using two methods: the chi-squared test and maximum error reference (Table 3). The IDW_4.0 model demonstrated the best performance in predicting WPM occurrence areas among the six spatial interpolation models. In the chi-squared test, IDW_4.0 showed $\chi^2 = 9.57$ with probability = 0.99, suggesting no significant difference between predicted and actual occurrence area values. In addition, IDW_4.0 achieved the highest prediction accuracy (84.04%) using the maximum error reference method among six spatial interpolation models. In summary, the IDW_4.0 model had the greatest potential to predict the WPM occurrence area among six spatial interpolation models.

4 DISCUSSION

The KNN, DT, RF, AdaBoost, SVM, and ANN algorithms were selected for predicting WPM severity degree because of their strengths in handling high-dimensional data, modeling nonlinear interactions, generalizing effectively and resisting overfitting, among other advantages. In this study, prediction models were constructed using each of six machine learning methods. To enhance model performance, we optimized their hyper-parameters, which significantly improved predictive accuracy. Among all models, SVM and KNN consistently demonstrated superior precision and accuracy across nearly all meteorological variable data sets and under cross-validation. The cross-validation results further confirmed the reliability and generalizability of the machine learning models.[15,40]

Notably, the SVM model trained on data set 3 achieved the best overall performance, with an accuracy of 0.76 and a macro-precision of 0.81. Alternatively, when using the data set 9, the KNN model performed best (accuracy: 0.68; macro-precision: 0.82). Additional metrics including macro-recall and macro

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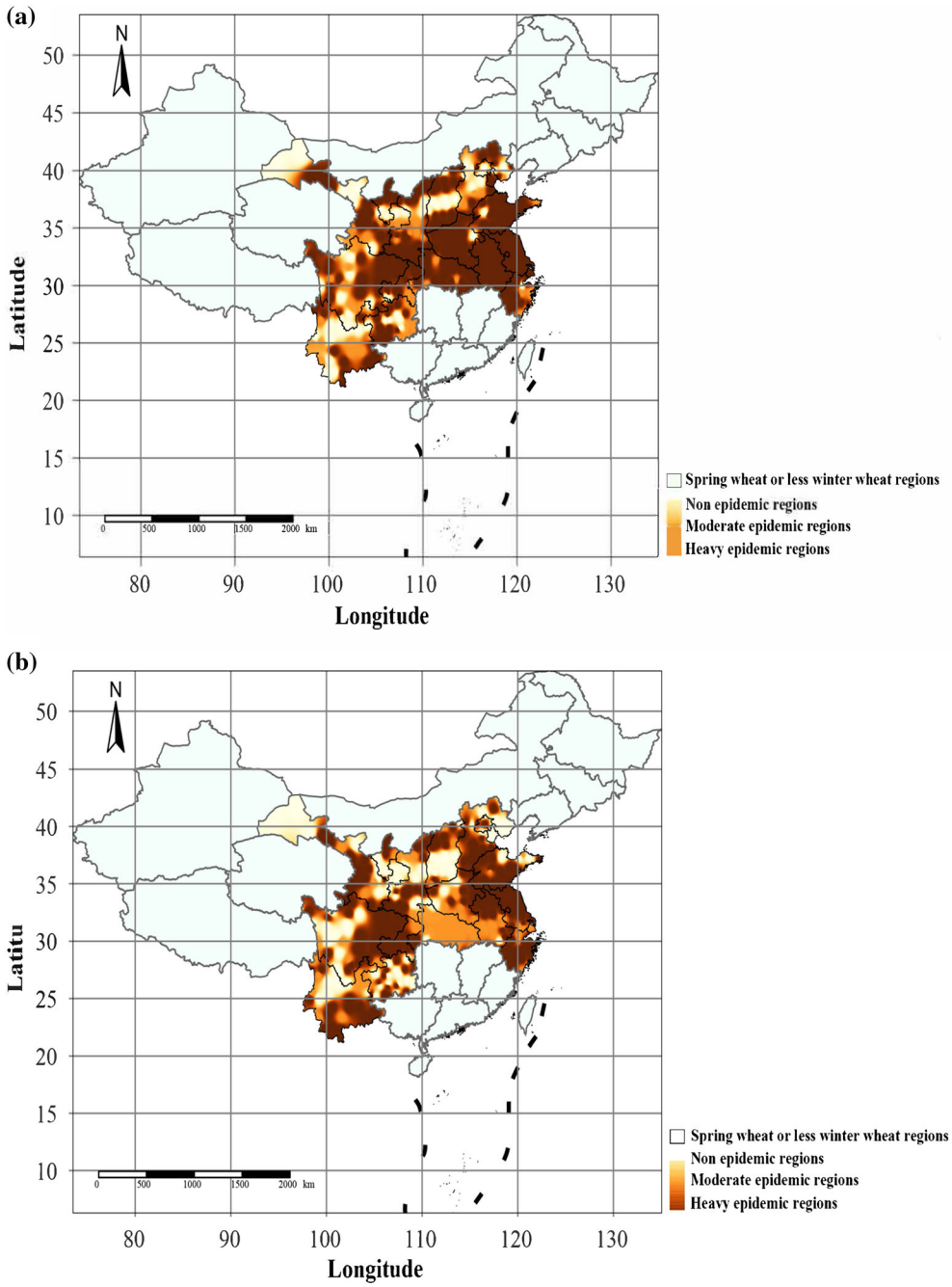


Figure 7. The disease distributions of wheat powdery mildew in China in 1991 (a) and 2002 (b), as spatial interpolated method using inverse distance weighted _4.0.

F1-score (provided in Supporting Information, Tables S4 and S5) also indicated that different machine learning algorithms were suited to different data configurations.[41–43]

Two major goals in the study of biological systems are inference and prediction. Inference creates a statistical model of the data-generation process to formalize understanding or test a hypothesis of how the system behaves. Machine learning algorithms aim to predict unobserved outcomes or future behavior. Machine learning algorithms make it possible to identify the best courses of action without requiring an understanding of the underlying mechanisms.[44] Inference involves building statistical models to formalize understanding or test hypotheses about system behavior, whereas machine learning focuses on forecasting unobserved or future outcomes. Although machine learning enables data-driven decision-making without requiring mechanistic understanding, it often sacrifices interpretability. For instance, Nie *et al.* developed region-specific statistical models across China, which improved predictive performance and offered insights into underlying mechanisms.[13] By contrast, machine learning models provide broader applicability—a single model can predict WPM severity degree nationwide—yet they remain less interpretable. Nevertheless, for practical deployment, machine learning-based predictive models outperform statistical approaches in terms of generalization and operational utility.

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Table 2. Actual and predicted occurrence areas of wheat powdery mildew by six different models (IDW_1.0, IDW_2.0, IDW_3.0, IDW_4.0, OK_exe and OK_pow) during 1990–2019

	Predicted occurrence area of WPM per million ha						
Year	IDW_1	IDW_2	IDW_3	IDW_4	OK_exc	OK_pow	Actual occurrence area of WPM per million ha
1990	7.54	7.44	9.95	7.58	7.62	7.49	12.08
1991	7.67	8.17	12.66	8.38	8.52	8.20	12.28
1992	7.64	7.87	11.27	7.99	8.10	7.73	7.59
1993	7.65	7.65	10.63	7.80	7.81	7.62	6.25
1994	7.32	7.13	8.03	7.04	7.03	7.00	6.21
1995	6.73	6.74	7.05	6.67	6.54	6.70	4.62
1996	7.61	7.83	11.27	7.96	7.93	7.75	5.62
1997	7.54	7.75	11.63	8.06	—	—	7.98
1998	7.50	7.45	9.25	7.31	7.42	7.48	7.25
1999	7.50	7.36	9.75	7.53	7.48	7.39	8.53
2000	6.69	6.42	6.04	6.41	6.30	6.45	6.00
2001	7.28	7.52	9.70	7.45	—	7.45	6.42
2002	7.17	7.29	8.94	7.24	7.25	7.26	6.70
2003	7.03	7.23	8.89	7.26	7.24	7.21	6.67
2004	6.69	6.73	7.35	6.81	—	—	5.90
2005	6.50	6.63	6.70	6.59	6.27	6.86	6.91
2006	6.35	5.67	2.98	5.48	5.41	5.82	6.70
2007	6.12	5.92	5.29	6.17	6.16	6.44	7.30
2008	7.19	6.67	6.66	6.57	6.55	6.63	6.53
2009	6.48	6.67	7.02	6.69	—	—	7.93
2010	6.26	6.66	-	6.51	6.55	5.18	6.90
2011	5.02	5.31	2.12	5.23	5.09	5.53	5.15
2012	7.22	6.87	7.45	6.83	6.82	6.83	5.40
2013	7.24	6.92	7.54	6.84	6.85	6.91	5.56
2014	7.24	7.29	8.55	7.14	7.11	7.17	5.70
2015	7.06	7.30	9.45	7.46	—	7.39	8.30
2016	7.14	6.76	6.80	6.57	6.62	6.80	7.86
2017	7.25	7.67	10.79	7.83	7.91	7.75	6.00
2018	7.21	7.08	7.67	6.79	6.98	7.10	6.00
2019	7.19	7.05	7.83	6.86	6.96	7.07	8.70

IDW, inverse distance weighted; OK, ordinary kriging; WPM, wheat powdery mildew; —, missing data.

Table 3. The validation results of values of the actual and predicted occurrence areas of wheat powdery mildew by chi-squared test and maximum error reference methods for six different models (IDW_1.0, IDW_2.0, IDW_3.0, IDW_4.0, OK_exe and OK_pow)

Models	χ^2	*P*-value	Accuracy (%)
IDW_1	14.31	0.9780401	75.91
IDW_2	10.35	0.9983739	82.08
IDW_3	9.56	0.9986291	83.39
IDW_4	9.57	0.9991968	84.04
OK_exc	8.78	0.9944032	83.10
OK_pow	10.23	0.9935315	82.45

IDW, inverse distance weighted; OK, ordinary kriging; χ^2, value of chi-squared test; accuracy, accuracy for predicting disease occurrence area with maximum error reference method.

Multicollinearity is indeed common in time-series meteorological data sets and can affect traditional statistical models such as linear regression. However, its impact on machine learning models is generally limited because: (i) many machine learning models do not heavily rely on interpreting individual feature weights, reducing concern over multicollinearity; (ii) tree-based models, for example, select the most informative feature among correlated variables during the split, which minimizes the effect of multicollinearity on predictive performance; and (iii) the primary goal of machine learning is predictive accuracy rather than interpretability.[45] Given these points, this study initially established the WPM severity degree model using the full data set with both 5-day

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and 10-day averages (Fig. 1(a)). Among the tested algorithms, the SVM model based on data set 3 (which includes 5-day interval meteorological data) demonstrated the best prediction performance. To address multicollinearity concerns, we also performed variable importance ranking using an RF forest algorithm (Fig. 1(b)), selected the top eight climate variables with minimal collinearity, and constructed a reduced data set. The KNN model on this reduced data set achieved slightly lower but still acceptable accuracy (0.68 *versus* 0.76 for SVM on full data), offering a practical balance between performance and variable redundancy.

The spatial distribution of WPM is influenced by geographic and climatic similarity—regions with comparable geographic conditions often exhibit similar disease severity degree. Both KNN and SVM are well-suited to capturing such spatial patterns: KNN leverages spatial autocorrelation by assigning similar predictions to nearby locations, whereas SVM delineates class boundaries using support vectors, making them effective for modeling WPM spread. Furthermore, because the relationships between meteorological factors and disease epidemics are complex and nonlinear, the ability of KNN and SVM to model such nonlinear interactions further supports their strong performance. In the future, we need to collect more data to improve the range and accuracy of prediction models, and explore the possibility of developing new models, such as fusing image and sensor data, lightweight models and complex methods (deep learning, hybrid models).[46–48]

Previous research has shown that epidemics of WPM were associated with climate variables in the winter and early spring periods.[9,11,20,21,49,50] The WPM epidemics were associated with the overwintering survival of *Bgt*, which is highly correlated with meteorological factors of the coldest month. In early spring, we mainly chose meteorological factors in the month before the wheat heading stage. Therefore, we chose meteorological factors for these two critical periods of the coldest month in winter (wheat overwintering stage) and the month before the wheat heading stage (wheat jointing–heading stage) in early spring in building the models. The period preceding the wheat heading stage is a critical window for WPM control—to forecast seasonal disease occurrence. This approach enables timely and informed decision-making for disease management by utilizing the latest variable—mean temperature 6–10 days before heading—enabling a forecast 5-day prior. With modern 10-day weather forecasts, an initial prediction is feasible 15 days ahead, providing practical support for both governmental guidance and farmer interventions. However, the climate variables in other wheat growth stages might affect the prediction of WPM severity degree,[51] which will be explored in further research. The RF variables' importance ranking showed that temperature and relative humidity in two main wheat growth stages were the most important variables for predicting the severity degree of WPM.

WPM prediction models for the prediction of WPM in 48 counties (cites) of 15 provinces in China were built using machine learning algorithms. The disease data were fairly representative because they were mainly from regions with winter WPM epidemics; however, we also need to collect more data to improve the range and accuracy of the prediction models. In this study, the original five-class severity data for WPM were redefined into three epidemic degrees—low, moderate and high—to reduce data complexity and enhance model practicality. This simplified classification aligns with common machine learning-based disease severity frameworks and better serves operational forecasting needs. Future work will explore alternative categorization methods to further improve model applicability.

Most published statistical models, and those in this study, did not consider the resistance of wheat varieties to WPM because most commercial wheat varieties are susceptible to WPM in China.[12,13] However, wheat varietal resistance to WPM was very important in reducing epidemics of WPM. The inoculum density of *Bgt* in fields was also a very important variable for epidemics of WPM. We will consider wheat varietal resistance to WPM and the density of *Bgt* in our models in further work. Epidemics of different plant diseases vary greatly, and our model is only for predicting epidemics of WPM; it may not be used for other diseases. We will explore the suitability of various models for prediction of different plant diseases in the future.

Previous research has shown that the severity of WPM has strong spatial dependency.[52] In this study, we used the wheat growth stage from different locations to represent regional characteristics. In future work, we should refine the location information to validate the model, and also plan to optimize both the severity degree prediction model and the spatial interpolation methods to improve the accuracy of occurrence area predictions.[53]

5 CONCLUSIONS

This study established WPM severity degree prediction models based on six machine learning algorithms with different meteorological variables and disease severity degree data sets (data set 1–6) throughout China. The best prediction model was based on the SVM algorithm with data set 3. The climate variables' importance ranking in data set 3, based on the RF method, indicated that X_{41}, X_4, X_{49}, X_8, X_{45}, X_{58}, X_{10}, X_{50} and X_{37} were the top eight most important variables for predicting WPM severity degree. Based on these findings, a KNN model incorporating data set 9 (these key variables and disease severity degrees) was selected for severity degree prediction because of its balanced performance and efficiency. Nationwide severity distributions were produced through spatial interpolation, enabling visual assessment of disease distributions and calculation of disease occurrence areas. Thus, further work may be focused on improving the accuracy of predicting WPM severity degree with multisource data and producing disease distributions with different spatial interpolators.

ACKNOWLEDGEMENTS

This research was financially supported by the National Key R&D Program of China (2022YFD1400400) and the National Natural Science Foundation of China (32072359).

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

SUPPORTING INFORMATION

Supporting information may be found in the online version of this article.

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